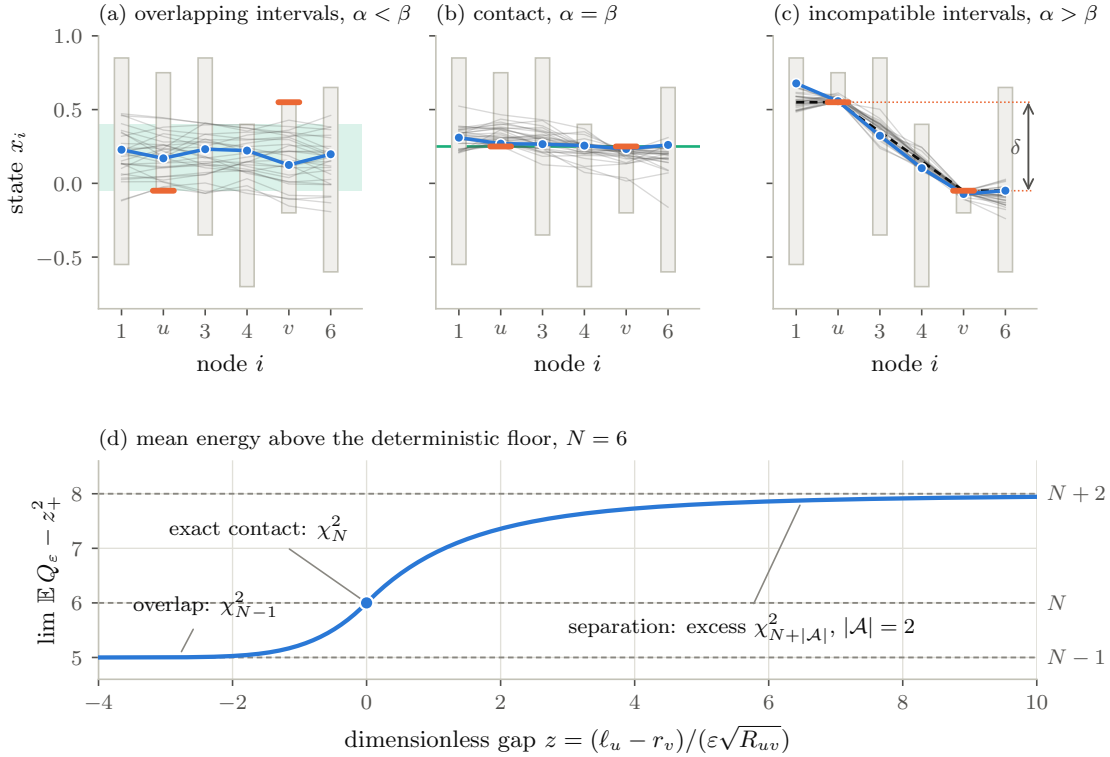

(a) overlapping intervals, $\alpha < \beta$
(b) contact, $\alpha = \beta$
(c) incompatible intervals, $\alpha > \beta$
state x_i
1.0
0.5
0.0
−0.5
1
u
3
4
v
6
node i
δ
(d) mean energy above the deterministic floor, $N = 6$
$\lim \mathbb{E}\, Q_\varepsilon - z_+^2$
8
7
6
5
$N + 2$
N
$N - 1$
exact contact: χ^2_N
overlap: χ^2_{N-1}
separation: excess $\chi^2_{N+|\mathcal{A}|}$, $|\mathcal{A}| = 2$
−4
−2
0
2
4
6
8
10
dimensionless gap $z = (\ell_u - r_v)/(\varepsilon\sqrt{R_{uv}})$